```
[swaraj@Swaraj ~]$ cd hadoop-test
[swaraj@Swaraj hadoop-test]$ ls
input  output
[swaraj@Swaraj hadoop-test]$ rm -rf output
[swaraj@Swaraj hadoop-test]$ ls
input
[swaraj@Swaraj hadoop-test]$ cd input
[swaraj@Swaraj input]$ ls
file1.txt  file2.txt
[swaraj@Swaraj input]$ cat file.txt
cat: file.txt: No such file or directory
[swaraj@Swaraj input]$ cat ./file.txt
cat: ./file.txt: No such file or directory
[swaraj@Swaraj input]$ cat ./file1.txt

hadoop mapreduce test
[swaraj@Swaraj input]$ cat ./file2.txt

learning hadoop is fun
[swaraj@Swaraj input]$ cd ..
[swaraj@Swaraj hadoop-test]$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/h
2026-04-18 21:19:35,624 INFO impl.MetricsConfig: Loaded properties from hadoo
2026-04-18 21:19:35,693 INFO impl.MetricsSystemImpl: Scheduled Metric snapsho
2026-04-18 21:19:35,693 INFO impl.MetricsSystemImpl: JobTracker metrics syste
2026-04-18 21:19:35,818 INFO input.FileInputFormat: Total input files to proc
2026-04-18 21:19:35,834 INFO mapreduce.JobSubmitter: number of splits:2
2026-04-18 21:19:35,950 INFO mapreduce.JobSubmitter: Submitting tokens for jo
2026-04-18 21:19:35,950 INFO mapreduce.JobSubmitter: Executing with tokens: [
2026-04-18 21:19:36,064 INFO mapreduce.Job: The url to track the job: http://
2026-04-18 21:19:36,067 INFO mapreduce.Job: Running job: job_local866432252_0
2026-04-18 21:19:36,070 INFO mapred.LocalJobRunner: OutputCommitter set in co
2026-04-18 21:19:36,079 INFO output.PathOutputCommitterFactory: No output com
2026-04-18 21:19:36,081 INFO output.FileOutputCommitter: File Output Committe
2026-04-18 21:19:36,081 INFO output.FileOutputCommitter: FileOutputCommitter
2026-04-18 21:19:36,082 INFO mapred.LocalJobRunner: OutputCommitter is org.ap
2026-04-18 21:19:36,135 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-18 21:19:36,136 INFO mapred.LocalJobRunner: Starting task: attempt_lo
2026-04-18 21:19:36,161 INFO output.PathOutputCommitterFactory: No output com
2026-04-18 21:19:36,161 INFO output.FileOutputCommitter: File Output Committe
2026-04-18 21:19:36,161 INFO output.FileOutputCommitter: FileOutputCommitter
2026-04-18 21:19:36,180 INFO mapred.Task:  Using ResourceCalculatorProcessTre
2026-04-18 21:19:36,184 INFO mapred.MapTask: Processing split: file:/home/swa
2026-04-18 21:19:36,219 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(1048575
2026-04-18 21:19:36,219 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-04-18 21:19:36,220 INFO mapred.MapTask: soft limit at 83886080
2026-04-18 21:19:36,220 INFO mapred.MapTask: bufstart = 0; bufvoid = 10485760
2026-04-18 21:19:36,220 INFO mapred.MapTask: kvstart = 26214396; length = 655
2026-04-18 21:19:36,227 INFO mapred.MapTask: Map output collector class = org
```

```
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=2
                Map output records=7
                Map output bytes=73
                Map output materialized bytes=99
                Input split bytes=220
                Combine input records=7
                Combine output records=7
                Reduce input groups=6
                Reduce shuffle bytes=99
                Reduce input records=7
                Reduce output records=6
                Spilled Records=14
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=908066816
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=45
        File Output Format Counters
                Bytes Written=62
[swaraj@Swaraj hadoop-test]$ ls
input  output
[swaraj@Swaraj hadoop-test]$ cat output/part-r-00000
fun     1
hadoop  2
is      1
learning        1
mapreduce       1
test    1
[swaraj@Swaraj hadoop-test]$
```

```
[swaraj@Swaraj log-mapreduce]$ ls
input  mapper.py  reducer.py
[swaraj@Swaraj log-mapreduce]$ cd input
[swaraj@Swaraj input]$ ls
log.txt
[swaraj@Swaraj input]$ cd ..
[swaraj@Swaraj log-mapreduce]$ nano mapper.py
[swaraj@Swaraj log-mapreduce]$ cat mapper.py
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    parts = line.strip().split()
    if len(parts) > 2:
        log_level = parts[2]
        print(f"{log_level}\t1")
[swaraj@Swaraj log-mapreduce]$ cat reducer.py
#!/usr/bin/env python3
import sys

current_key = None
current_count = 0

for line in sys.stdin:
    key, value = line.strip().split("\t")
    value = int(value)

    if key == current_key:
        current_count += value
    else:
        if current_key:
            print(f"{current_key}\t{current_count}")
        current_key = key
        current_count = value

if current_key:
    print(f"{current_key}\t{current_count}")
[swaraj@Swaraj log-mapreduce]$ hadoop jar $HADOOP_HOME/share/hadoop/tools/
-input input/log.txt \
-output output \
-mapper mapper.py \
-reducer reducer.py
2026-04-18 21:29:50,824 INFO impl.MetricsConfig: Loaded properties from ha
2026-04-18 21:29:50,889 INFO impl.MetricsSystemImpl: Scheduled Metric snap
2026-04-18 21:29:50,889 INFO impl.MetricsSystemImpl: JobTracker metrics sy
2026-04-18 21:29:50,898 WARN impl.MetricsSystemImpl: JobTracker metrics sy
2026-04-18 21:29:51,004 INFO mapred.FileInputFormat: Total input files to
```

```
2026-04-18 21:29:51,550 INFO mapred.LocalJobRunner: Finishing task: attempt_local
2026-04-18 21:29:51,550 INFO mapred.LocalJobRunner: reduce task executor complete
2026-04-18 21:29:52,244 INFO mapreduce.Job: Job job_local2105644984_0001 running
2026-04-18 21:29:52,245 INFO mapreduce.Job:  map 100% reduce 100%
2026-04-18 21:29:52,246 INFO mapreduce.Job: Job job_local2105644984_0001 complete
2026-04-18 21:29:52,255 INFO mapreduce.Job: Counters: 30
        File System Counters
                FILE: Number of bytes read=292004
                FILE: Number of bytes written=1575160
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=78
                Map output records=78
                Map output bytes=626
                Map output materialized bytes=788
                Input split bytes=97
                Combine input records=0
                Combine output records=0
                Reduce input groups=3
                Reduce shuffle bytes=788
                Reduce input records=78
                Reduce output records=3
                Spilled Records=156
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=7
                Total committed heap usage (bytes)=685768704
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=3778
        File Output Format Counters
                Bytes Written=40
2026-04-18 21:29:52,255 INFO streaming.StreamJob: Output directory: output
[swaraj@Swaraj log-mapreduce]$ cat output/part-00000
ERROR   29
INFO    32
WARNING 17
[swaraj@Swaraj log-mapreduce]$
```